Calculus II Lesson 34: Conics in Polar Coordinates

Polar Equations of Conics:

$$r = \frac{ed}{1 \pm e\cos\theta} \quad \text{or} \quad r = \frac{ed}{1 \pm e\sin\theta}$$

where e (eccentricy) > 0 and

$|d| =$ distance between the focus at the pole and directrix.

Note: Focus is at the pole.

Classification of Conics by Eccentricity

Let $e =$ eccentricity.

a) The conic is a ellipse for $0 < e < 1$.

b) The conic is a parabola for $e = 1$.

c) The conic is a hyperbola for $e > 1$.

Rectangular and Polar Systems

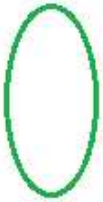
$$x = r\cos\theta; \quad y = r\sin\theta$$

$$x^2 + y^2 = r^2; \quad r = \pm\sqrt{x^2 + y^2}$$

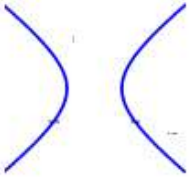
$$\tan\theta = \frac{y}{x}; \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

polar equation for the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $r^2=\frac{b^2}{1-e^2\cos^2\theta}$.

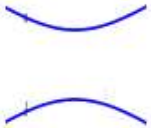
polar equation for the ellipse $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$ is $r^2=\frac{b^2}{1-e^2\sin^2\theta}$.



polar equation for the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $r^2=\frac{-b^2}{1-e^2\cos^2\theta}$



polar equation for the hyperbola $\frac{y^2}{a^2}-\frac{x^2}{b^2}=1$ is $r^2=\frac{-b^2}{1-e^2\sin^2\theta}$



1) $r = \dfrac{4}{1 - \cos\theta}$

a) Graph $r = \dfrac{4}{1 - \cos\theta}$ (Multiple choice.) Choose the correct answer (blue, red, green, or orange)

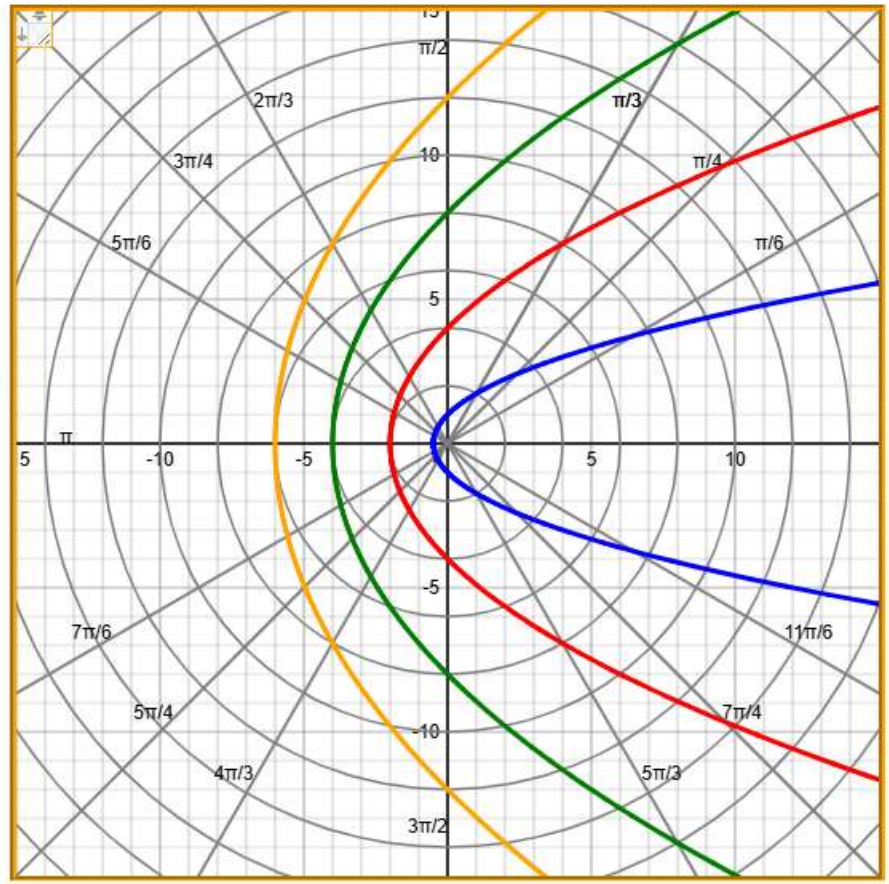
b) eccentricity = ___?___

c) type of conic section = ___?___ (parabola, ellipse, or hyperbola)

d) Write the corresponding rectangular equation for $r = \dfrac{3}{1-\cos\theta}$.

Hint: Multiply both sides by $1-\cos\theta$.

$y^2 =$ _____?_____



Answers

a) b) c) d)

2) Polar equation $r = \dfrac{4}{5 - 2\cos\theta}$.

a) Graph $r = \dfrac{4}{5 - 2\cos\theta}$. (Multiple choice.) Choose the correct answer (blue, red, green, or orange)

b) Multiplying by $\dfrac{1}{5}$: $r = \dfrac{(1/5)\cdot 4}{(1/5)\cdot 5 - (1/5)\cdot 2\cos\theta} =$ ___?___

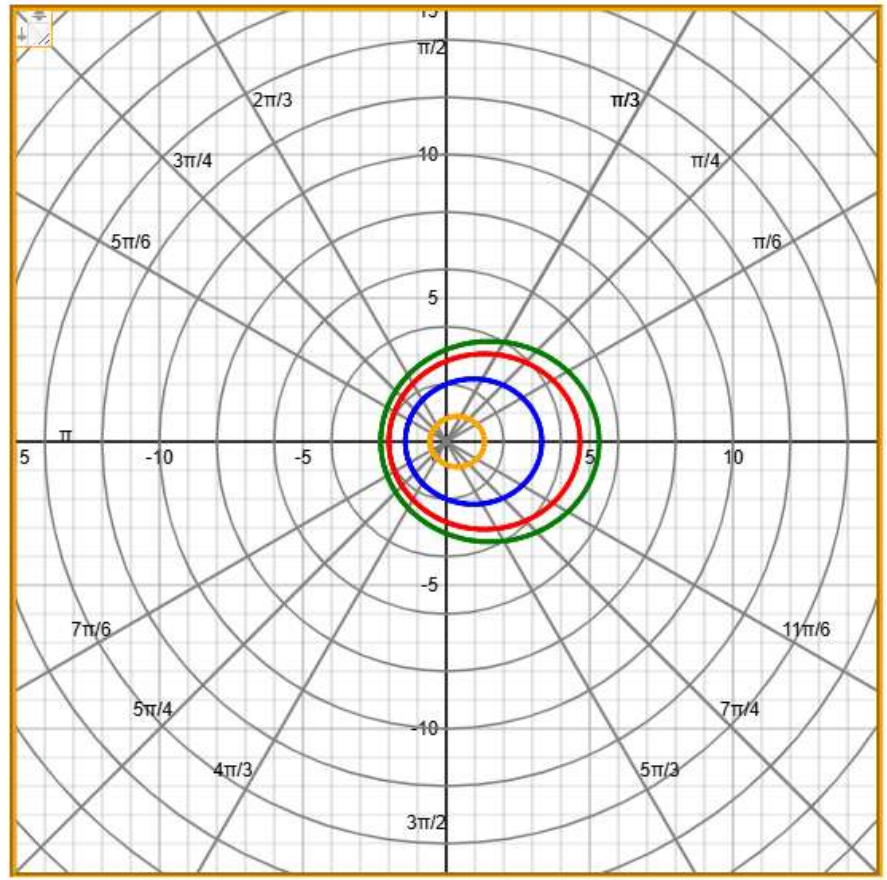
c) eccentricity = ___?___

d) Write the corresponding rectangular equation.

Hint: Multiply both sides of $r = \dfrac{4}{5 - 2\cos\theta}$ by $5 - 2\cos\theta$.

Use $r = \sqrt{x^2 + y^2}$; $x = r\cos\theta$; $y = r\sin\theta$

$25x^2 + 25y^2 =$ _____?_____



Answers

a) b) c) d)

3) Polar equation $r = \dfrac{5}{-1 + 4\cos\theta}$.

a) Graph $r = \dfrac{5}{-1 + 4\cos\theta}$. (multiple choice) Choose the correct answer (blue, red, green, or yellow)

b) Multiplying by -1: $r = \dfrac{(-1)\cdot 5}{(-1)\cdot(-1) + (-1)\cdot\ 4\cos\theta} =$ ___?___

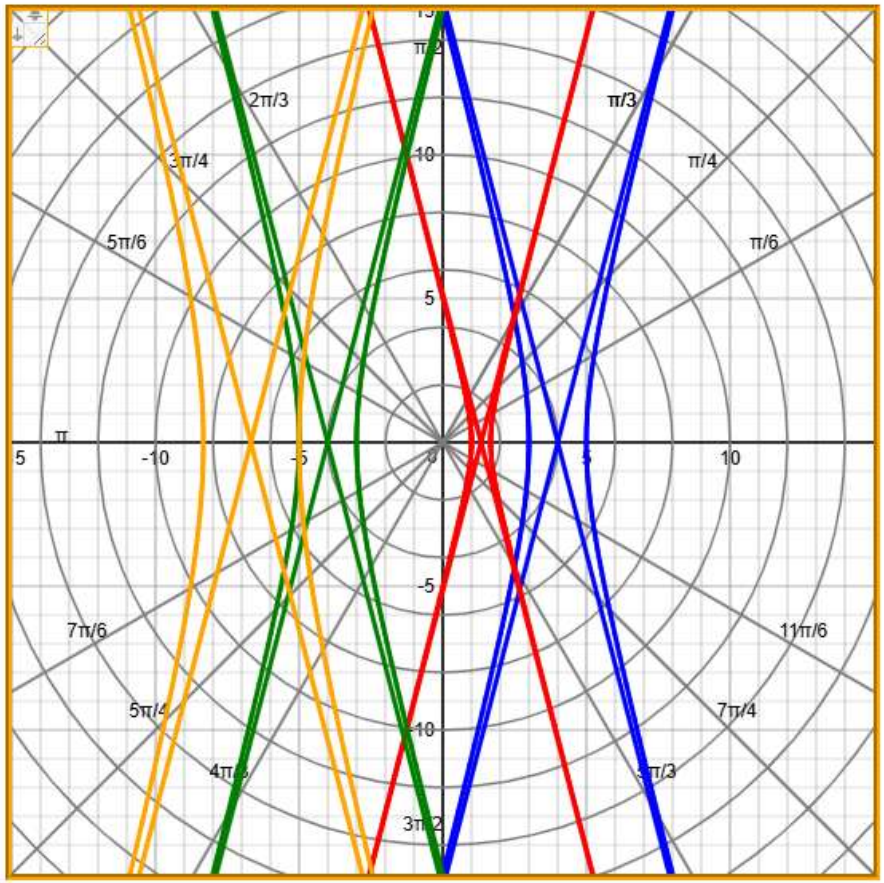
c) eccentricity = ___?___

d) Write the corresponding rectangular equation.

Hint: Multiply both sides of $r = \dfrac{5}{-1 + 4\cos\theta}$ by $-1 + 4\cos\theta$.

Use $r = \sqrt{x^2 + y^2}$; $x = r\cos\theta$; $y = r\sin\theta$

$x^2 + y^2 =$ _____?_____



Answers

a) b) c) d)

4) Find a polar equation for the parabola with its focus at the pole.

Parabola has: $e = \text{eccentricity} = 1$; Directrix: $x = -2$

a) focus = _____

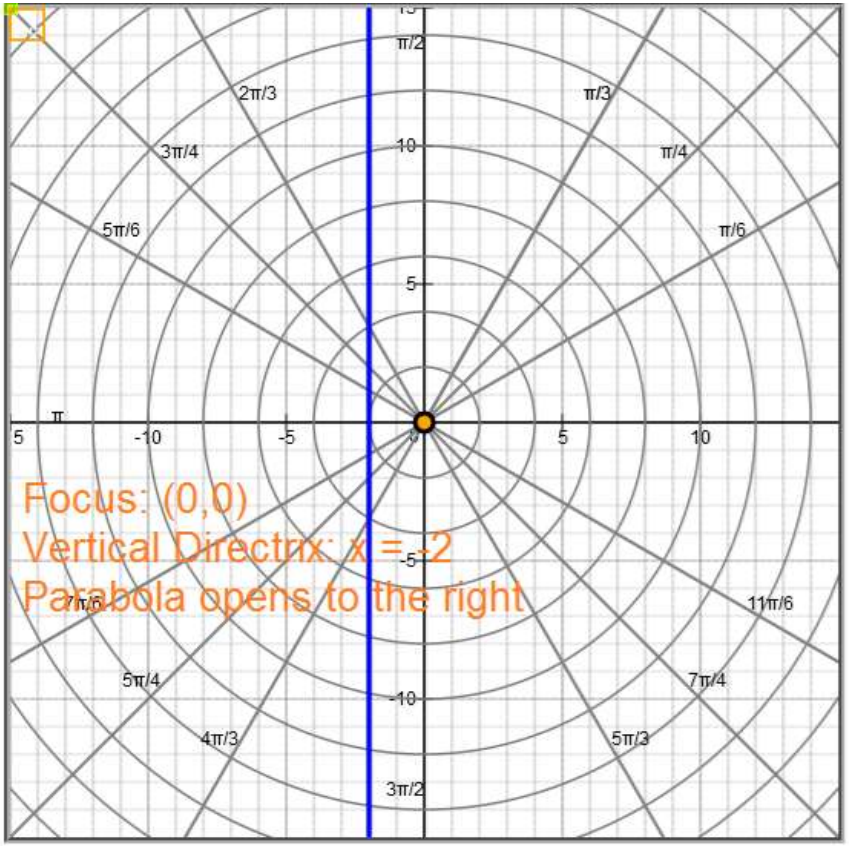
b) vertex = _____

c) Find distance from focus at the pole to directrix $= |d| =$ ___?___

d) Polar equation for parabola: ___?___ (multiple choice)

A) $r = \dfrac{2}{1+\cos\theta}$ B) $r = \dfrac{2}{1-\cos\theta}$

C) $r = \dfrac{-2}{1+\cos\theta}$ D) $r = \dfrac{-2}{1-\cos\theta}$



Answers

a) b) c) d)

5) Find a polar equation for the parabola with its focus at the pole. Parabola has vertex at $(1,\ \pi/2)$.

a) focus = _____ b) Equation of directrix: _____

c) Find distance from focus at the pole to directrix $=|d|=$ ___?___

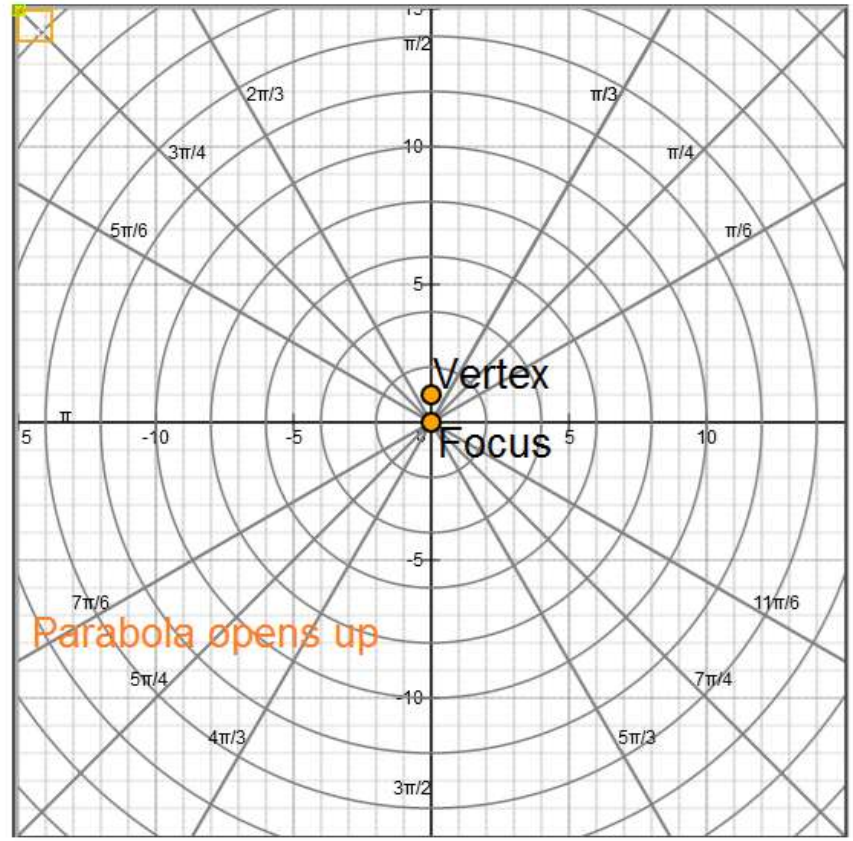
d) e = eccentricity = _____

Hint: To find e, use the equation $r=\dfrac{ed}{1\pm e\sin\theta}$ and the point $(r=1,\ \theta=\pi/2)$.

e) Polar equation for parabola: ___?___ (multiple choice)

A) $r=\dfrac{1}{1+\sin\theta}$ B) $r=\dfrac{2}{1+\sin\theta}$

C) $r=\dfrac{3}{1-\sin\theta}$ D) $r=\dfrac{4}{1-\sin\theta}$



Answers

a) b) c) d) e)

6) Find a polar equation for the ellipse with its focus at the pole.

Ellipse has: $e = \text{eccentricity} = 1/3$; Directrix: $y = 3$

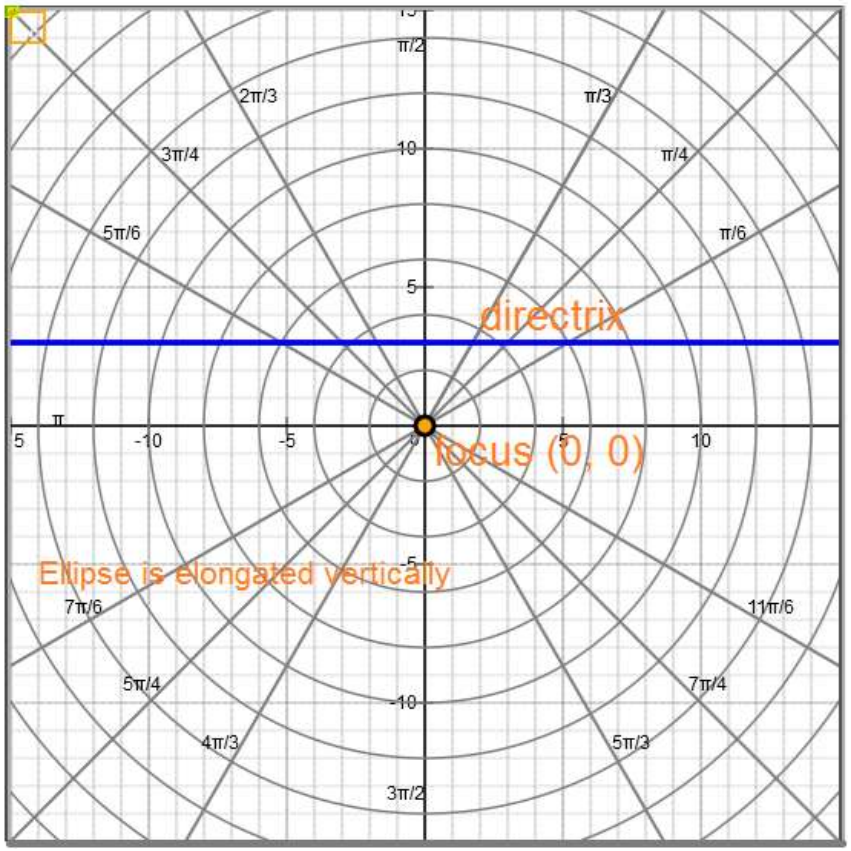
a) focus =_____

b) Find distance from focus at the pole to directrix $= |d| =$ ___?___

c) Polar equation for ellipse: ___?___

Equation of ellipse will be: $r = \dfrac{ed}{1 + e\sin\theta}$ or $r = \dfrac{ed}{1 - e\sin\theta}$.

Note: One of the vertices of the ellipse is closer to the pole.
Directrix should be near the vertex closest to the pole.



Answers

a) b) c)

7) Find a polar equation for the ellipse with its focus at the pole.
Ellipse has vertices at (2, 0) and (6, π).

a) focus = ___?___

b) center of ellipse = ___?___

c) $a =$ distance from center to vertex = ___?___

d) $c =$ distance from center to focus = ___?___

e) $e =$ eccentricity $= c/a =$ ___?___

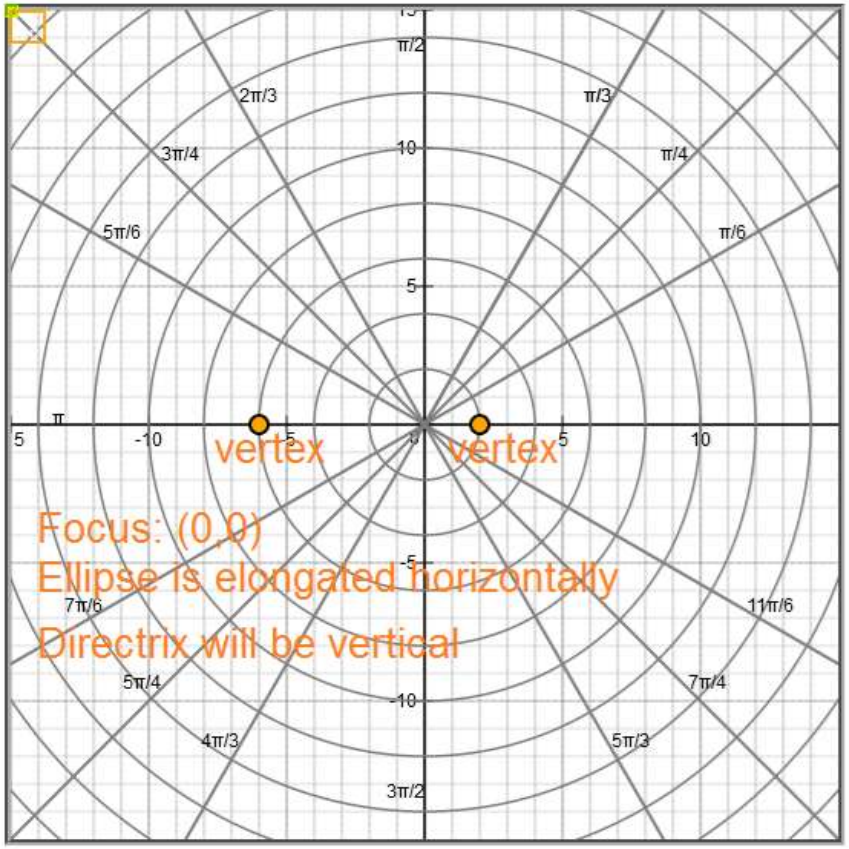
f) Find distance from focus at the pole to directrix $= |d| =$ ___?___

Hint: Equation of ellipse will have the form: $r = \dfrac{ed}{1 \pm e\cos\theta}$

Use the point ($r = 2$, $\theta = 0$) to find d.

g) Polar equation for ellipse: ___?___

Note: One of the vertices of the ellipse is closer to the pole.
Directrix should be near the vertex closest to the pole.



Answers

a) b) c) d) e) f) g)

8) Find a polar equation for the hyperbola with its focus at the pole.

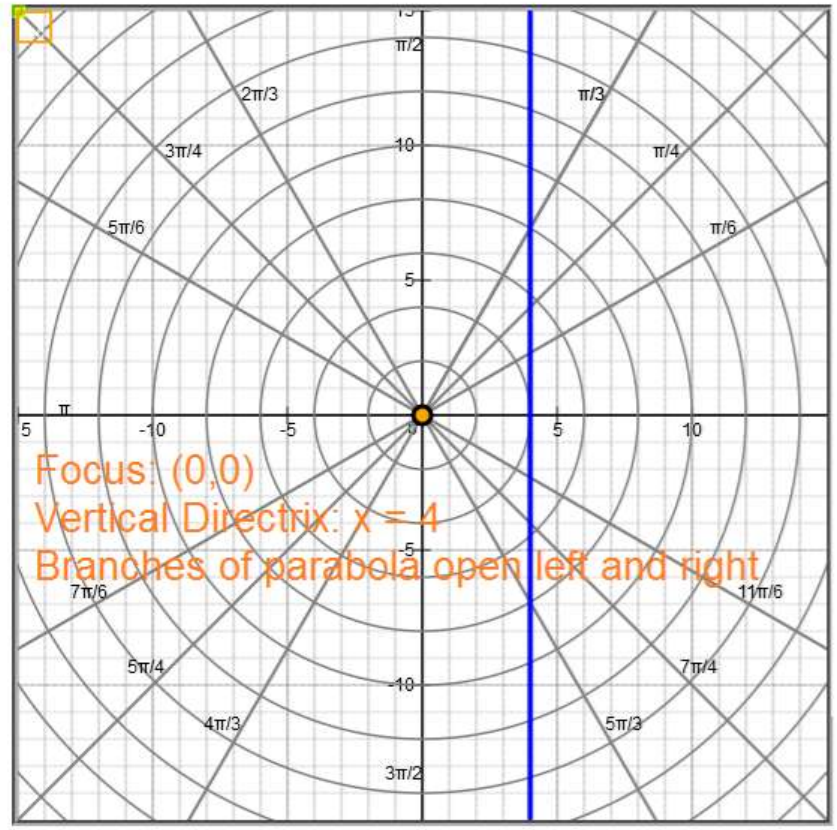
Hyperbola has: e = eccentricity = 4; Directrix: $x = 4$

a) Find distance from focus at the pole to directrix = d = ___?___

b) Polar equation for hyperbola: ___?___

Equation of hyperbola will be: $r = \dfrac{ed}{1+e\cos\theta}$ or $r = \dfrac{ed}{1-e\cos\theta}$.

Note: Directrix should be between the two branches of the hyperbola.



Answers

a) b) c) d) e) f) g)

9) Find a polar equation for the hyperbola with its focus at the pole.

Hyperbola has vertices at $(1,\ 3\pi/2)$ and $(7,\ 3\pi/2)$.

a) focus = ___?___

b) center of hyperbola = ___?___

c) $a =$ distance from center to vertex = ___?___

d) $c =$ distance from center to focus = ___?___

e) $e =$ eccentricity $= c/a =$ ___?___

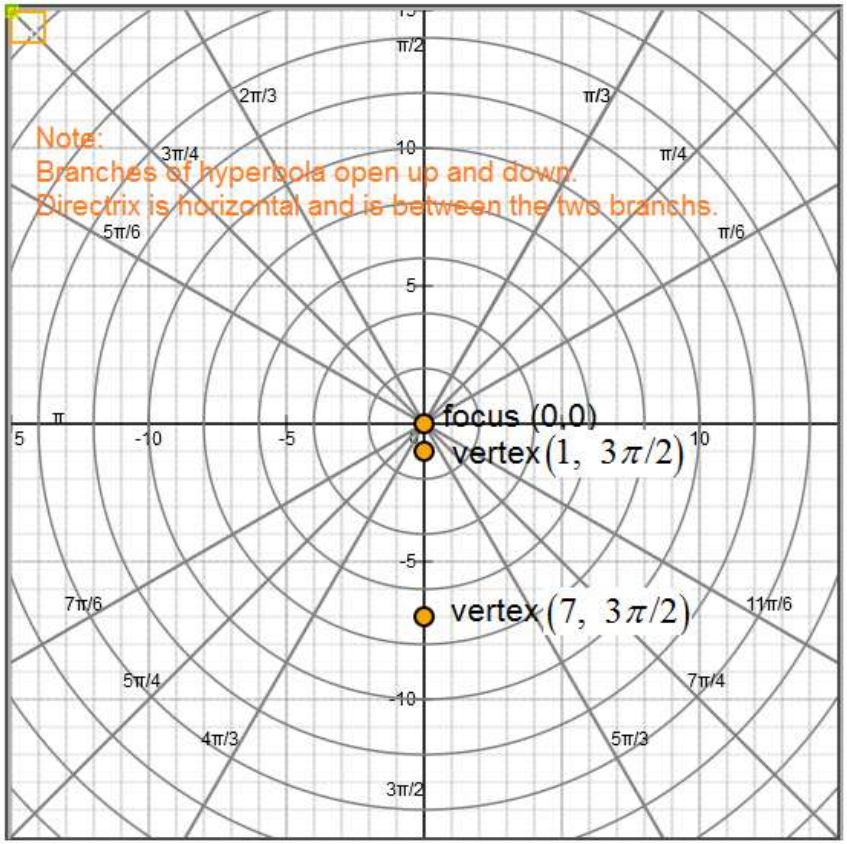
f) Find distance from focus at the pole to directrix $= |d| =$ ___?___

Hint: Equation of hyperbola will have the form: $r = \dfrac{ed}{1 \pm e\sin\theta}$.

Use the point $(r = 1,\ \theta = 3\pi/2)$ to find d.

g) Polar equation for hyperbola: ___?___

Note: Directrix should be between the two branches of the hyperbola.



Answers

a) b) c) d) e) f) g)